

WEST BENGAL JEE 2026

MATHEMATICS MOCK TEST QUESTION

General Instructions

Category-I: 1 Mark. Category-II: 2 Marks. Category-III: 2 Marks (Multiple Correct).

Category-I (Q1 - Q50)

Q1. The set of points of discontinuity of the function $f(x) = x - [x]$ for $x \in \mathbb{R}$ is:

- (A) $\mathbb{Q}$
- (B) $\mathbb{R}$
- (C) $\mathbb{N}$
- (D) $\mathbb{Z}$

Ans: (D)

Q2. Two circles $S_1 = px^2 + py^2 + 2g'x + 2f'y + d = 0$ and $S_2 = x^2 + y^2 + 2gx + 2fy + d' = 0$ have a common chord PQ. The equation of PQ is:

- (A) $S_1 - S_2 = 0$
- (B) $S_1 + S_2 = 0$
- (C) $S_1 - pS_2 = 0$
- (D) $S_1 + pS_2 = 0$

Ans: (C)

Q3. If the sum of the distances of a point from two perpendicular lines in a plane is 1 unit, then its locus is:

- (A) A square
- (B) A circle
- (C) A straight line
- (D) Two intersecting lines

Ans: (A)

Q4. Let $f(x) = |x^2 - 1|$, $x \in \mathbb{R}$ Then:

- (A) f has a local minimum at $x = \pm 1$ but no local maximum.
- (B) f has a local maximum at $x = 0$ but no local minimum.
- (C) f has a local minima at $x = \pm 1$ and a local maxima at $x = 0$.
- (D) f has neither a local maxima nor a local minima at any point.

Ans: (C)

Q5. The approximate value of $\sin 31^\circ$ is:

- (A) > 0.5
- (B) > 0.6
- (C) < 0.5
- (D) < 0.4

Ans: (A)

Q6. The limit of the interior angle of a regular polygon of n sides as $n \rightarrow \infty$ is:

- (A) π
- (B) $\pi/2$
- (C) $3\pi/2$
- (D) 0

Ans: (A)

Q7. The value of $\int [x] dx$ from 0 to n , where $n \in \mathbb{N}$ is:

- (A) n
- (B) $n - 1$
- (C) $n(n - 1) / 2$
- (D) $n(n + 1) / 2$

Ans: (C)

Q8. If a, b, c are real numbers each greater than 1 such that $\log_b(a) + \log_c(b) + \log_a(c) = 3$ and $b = 9$, then the value of a is:

- (A) 3
- (B) 81
- (C) 18
- (D) 27

Ans: (D)

Q9. Let $f(x)$ be a derivable function such that $f'(x) > f(x)$ and $f(0) = 0$. Then for all $x > 0$:

- (A) $f(x) > 0$
- (B) $f(x) < 0$
- (C) $f(x) = 0$
- (D) Sign uncertain

Ans: (A)

Q10. The number of reflexive relations on a set containing n elements is:

- (A) $2^{(n^2)}$
- (B) $2^{(n(n - 1))}$
- (C) $2^{(n^2 - n)}$
- (D) 2^n

Ans: (C)

Q11. If $\mathbf{a} = 3\mathbf{i} - \mathbf{k}$ and $|\mathbf{b}| = \sqrt{5}$ and $\mathbf{a} \cdot \mathbf{b} = 3$, then area of parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$ is:

- (A) $\sqrt{17}$
- (B) $\sqrt{14}$
- (C) $\sqrt{7}$
- (D) $\sqrt{41}$

Ans: (D)

Q12. If P is the extremity of the latus rectum of the ellipse $3x^2 + 4y^2 = 48$ in the first quadrant, the eccentric angle of P is:

- (A) $\pi/6$
- (B) $3\pi/4$
- (C) $\pi/3$
- (D) $2\pi/3$

Ans: (C)

Q13. The equation of the plane which bisects the line joining points (1, 2, 3) and (3, 4, 5) at right angles is:

- (A) $x + y + z = 0$
- (B) $x + y - z = 9$
- (C) $x + y + z = 9$
- (D) $x + y - z + 9 = 0$

Ans: (C)

Q14. If $f(x)$ and $h(x)$ are inverse functions and $h'(x) = 1 / (1 + \log x)$ then $f'(x)$ is:

- (A) $1 + \log(f(x))$
- (B) $1 + f(x)$
- (C) $1 - \log(f(x))$
- (D) $\log f(x)$

Ans: (A)

Q15. Number of quadratic equations with coefficients from $\{0,1,2,\dots,9\}$ such that 0 is a solution:

- (A) 29
- (B) 39
- (C) 90
- (D) 81

Ans: (C)

Q16. If a, b, c are distinct odd natural numbers, the number of rational roots of $ax^2 + bx + c = 0$ is:

- (A) 0
- (B) 1
- (C) 2
- (D) Infinite

Ans: (A)

Q17. The odds in favor of throwing at least 8 in a single throw with two dice are:

- (A) 5:13
- (B) 5:31
- (C) 5:7
- (D) 5:6

Ans: (C)

Q18. If $P(A \cup B) = 0.6$ and $P(A \cap B) = 0.2$, then $P(A) + P(B)$ is:

- (A) 1.8
- (B) 1.2
- (C) 0.6
- (D) 0.4

Ans: (B)

Q19. In a box of 100 bulbs (10 def), the prob that sample of 5 has none defective is:

- (A) 10^{-1}
- (B) $(1/2)^5$
- (C) $(9/10)^5$
- (D) $9/10$

Ans: (C)

Q20. If independent $P(A) = 1/2$ and $P(B) = 1/5$, then $P(A \cup B)$ is:

- (A) $3/10$
- (B) $1/10$
- (C) $6/10$
- (D) $7/10$

Ans: (C)

Q21. Point in circle interior. Prob point closer to center than boundary is:

- (A) $1/2$
- (B) $1/4$
- (C) $1/3$
- (D) $1/8$

Ans: (B)

Q22. If $X \sim \text{Bin}(100, 1/3)$, $P(X = r)$ is max when r is:

- (A) 32
- (B) 33
- (C) 34
- (D) 35

Ans: (B)

Q23. Value of det with log entries $|\begin{matrix} 1 & \log_a b & \log_a c \end{matrix}|$ etc. is:

- (A) 1
- (B) 0
- (C) e
- (D) $\log(abc)$

Ans: (B)

Q24. If $|A| = 6$ and $\text{adj } A = \begin{bmatrix} 1 & -2 & 4 \\ 4 & 1 & 1 \\ -1 & k & 0 \end{bmatrix}$, then k is:

- (A) -1
- (B) 1
- (C) 2
- (D) 0

Ans: (C)

Q25. If $x = -1, x = 2$ extreme points of $f(x) = \alpha \log|x| + \beta x^2 + x$ then:

- (A) $\alpha = -6, \beta = 1/2$
- (B) $\alpha = -6, \beta = -1/2$
- (C) $\alpha = 2, \beta = -1/2$
- (D) $\alpha = 2, \beta = 1/2$

Ans: (C)

Q26. Polynomial $p(x)$ with local max at $x=1$, local min at $x=3$, $p(1)=6$, $p(3)=2$. $p'(0)$ is:

- (A) 8
- (B) 9
- (C) 3
- (D) 6

Ans: (B)

Q27. If $x = \int \frac{1}{\sqrt{1 + 9t^2}} dt$ from 0 to y and $d^2y/dx^2 = ay$, then a is:

- (A) 3
- (B) 6
- (C) 9
- (D) 1

Ans: (C)

Q28. The value of $\int (x^3 + |x| + 1)/(x^2 + 2|x| + 1) dx$ from -1 to 1 is:

- (A) $\log 2$
- (B) $2 \log 2$
- (C) $\frac{1}{2} \log 2$
- (D) $4 \log 2$

Ans: (B)

Q29. If ${}^9P_5 + 5 \cdot {}^9P_4 = 10P_r$ then r is:

- (A) 4
- (B) 8
- (C) 5
- (D) 10

Ans: (C)

Q30. The value of $\int [x^2] dx$ from 0 to 1.5 is:

- (A) 2
- (B) $2 - \sqrt{2}$
- (C) $2 + \sqrt{2}$
- (D) 1.5

Ans: (B)

Q31. For $f(x) = 2x^3 - 3x^2 - 12x + 4$, $x \in \mathbb{R}$:

- (A) Two local maxima
- (B) Two local minima
- (C) One local max and one local min
- (D) Neither

Ans: (C)

Q32. The expression $2^{(4n)} - 15n - 1$ is divisible by:

- (A) 125
- (B) 225
- (C) 325
- (D) 425

Ans: (B)

Q33. Line $y - \sqrt{3}x + 3 = 0$ cuts $y^2 = x + 2$ at P, Q. If $X = (\sqrt{3}, 0)$, $XP \cdot XQ$ is:

- (A) $\frac{4}{3}(2\sqrt{3} + 1)$
- (B) $4(2\sqrt{3} - 1)$
- (C) $2\sqrt{3}$
- (D) $5(2\sqrt{3})/3$

Ans: (D)

Q34. If f inverse of g and $g'(x) = 1 / (1 + x^n)$ then $f'(x)$ is:

- (A) $1 + (f(x))^n$
- (B) $1 - (f(x))^n$
- (C) $(1 + f(x))^n$
- (D) $(f(x))^n$

Ans: (A)

Q35. Number of solutions of $2 \cos^2(x/2) \sin^2 x = x^2 + 1/x^2$ for $x \in (0, \pi/2)$ is:

- (A) Zero
- (B) 1
- (C) Infinite
- (D) None

Ans: (A)

Q36. Focus of parabola $x = 5t^2 + 2$, $y = 10t + 4$ is:

- (A) (7, 4)
- (B) (3, 4)
- (C) (3, -4)
- (D) (-7, 4)

Ans: (A)

Q37. In ΔABC if $\sin A \sin B = ab/c^2$ the triangle is:

- (A) Equilateral
- (B) Isosceles
- (C) Right angled
- (D) Obtuse angled

Ans: (C)

Q38. If a, b, c in G.P., then $\log a^n, \log b^n, \log c^n$ are in:

- (A) A.P.
- (B) G.P.
- (C) H.P.
- (D) None

Ans: (A)

Q39. The smallest value of $5 \cos \theta + 12$ is:

- (A) 5
- (B) 12
- (C) 7
- (D) 17

Ans: (C)

Q40. General solution of $dy/dx = e^{(y+x)} + e^{(y-x)}$ is:

- (A) $e^{-y} = e^x - e^{-x} + C$
- (B) $e^{-y} = e^{-x} - e^x + C$
- (C) $e^{-y} = e^x + e^{-x} + C$
- (D) $e^y = e^x + e^{-x} + C$

Ans: (B)

Q41. Product of any r consecutive natural numbers is always divisible by:

- (A) $r!$
- (B) $(r + 4)!$
- (C) $(r + 1)!$
- (D) $2r$

Ans: (A)

Q42. Number of common terms in 17, 21..417 and 16, 21..466 is:

- (A) 21
- (B) 19
- (C) 20
- (D) 91

Ans: (C)

Q43. Circle through (4, 5) with center (2, 2) is:

- (A) $x^2 + y^2 + 4x + 4y - 5 = 0$
- (B) $x^2 + y^2 - 4x - 4y - 5 = 0$
- (C) $x^2 + y^2 + 4x - 13 = 0$
- (D) $x^2 + y^2 - 4x - 4y + 5 = 0$

Ans: (B)

Q44. Common tangents to $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$ is:

- (A) 0
- (B) 1
- (C) 3
- (D) 4

Ans: (B)

Q45. If $\sin x + \sin^2 x = 1$, the value of $\cos^2 x + \cos^4 x$ is:

- (A) 1
- (B) 2
- (C) 1.5
- (D) 0

Ans: (A)

Q46. For every point $P(x,y,z)$ on the XY -plane:

- (A) $x = 0$
- (B) $y = 0$
- (C) $z = 0$
- (D) $x + y = 0$

Ans: (C)

Q47. In ΔABC if $AB = BC = 6$ cm and $\angle ABC = 1.8$ radians, area is:

- (A) $18 \sin(1.8)$
- (B) $36 \sin(1.8)$
- (C) $9 \sin(1.8)$
- (D) $18 \cos(1.8)$

Ans: (A)

Q48. If $2z_1/3z_2$ is purely imaginary, value of $|(z_1 - z_2)/(z_1 + z_2)|$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Ans: (A)

Q49. Third term in $(x^2 - 1/x^3)^n$ is independent of x when n is:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Ans: (D)

Q50. The differential of x^6 with respect to x^3 is:

- (A) $6x^3$
- (B) $3x^2$
- (C) $2x^3$
- (D) x^3

Ans: (C)

Category-II (Q51 - Q65)

Q51. Area bounded by $x = 4 - y^2$ and the Y -axis is:

- (A) 16
- (B) $32/3$
- (C) $16/3$
- (D) 32

Ans: (B)

Q52. If $S_n = 3n^2 + 5n$ and m-th term is 164, value of m is:

- (A) 26
- (B) 27
- (C) 28
- (D) 29

Ans: (B)

Q53. If $\cos^{-1} \alpha + \cos^{-1} \beta + \cos^{-1} \gamma = 3\pi$, then $\alpha(\beta + \gamma) + \beta(\gamma + \alpha) + \gamma(\alpha + \beta)$ is:

- (A) 0
- (B) 1
- (C) 6
- (D) 12

Ans: (C)

Q54. Value of $\int \sqrt{x} / (\sqrt{5-x} + \sqrt{x}) dx$ from 2 to 3 is:

- (A) 1
- (B) 0.5
- (C) 2
- (D) 1.5

Ans: (B)

Q55. Coefficient of x^5 in $(1+x)^{21} + \dots + (1+x)^{30}$ is:

- (A) $51C_5$
- (B) $9C_5$
- (C) $31C_6 - 21C_6$
- (D) $30C_5 + 20C_5$

Ans: (C)

Q56. If $\int 2^x / \sqrt{1-4^x} dx = (k / \log 2) \sin^{-1}(2^x) + C$, then k is:

- (A) $\log 2$
- (B) 0.5
- (C) 1
- (D) 2

Ans: (C)

Q57. Area bounded by $x + 2|y| = 1$ and $x = 0$ is:

- (A) $1/4$
- (B) $1/2$
- (C) 1
- (D) 2

Ans: (B)

Q58. If $(1 + x - 2x^2)^6 = 1 + a_1x + \dots + a_{12}x^{12}$, then $a_2 + a_4 + \dots + a_{12}$ is:

- (A) 21
- (B) 31
- (C) 32
- (D) 64

Ans: (B)

Q59. Minimum value of $|a + b\omega + c\omega^2|^2$ for distinct non-zero integers a, b, c is:

- (A) 15
- (B) 5
- (C) 3
- (D) 4

Ans: (C)

Q60. For $f(x) = |1 - 2x|$, function is:

- (A) Cont. not diff.
- (B) Diff. not cont.
- (C) Both
- (D) Neither

Ans: (A)

Q61. If matrix $\begin{bmatrix} 0 & a & a \\ 2b & b & -b \\ c & -c & c \end{bmatrix}$ is orthogonal, then a, b, c are:

- (A) $1/\sqrt{3}, 1/\sqrt{6}, 1/\sqrt{2}$
- (B) $1/\sqrt{2}, 1/\sqrt{6}, 1/\sqrt{3}$
- (C) $1/\sqrt{3}, 1/\sqrt{2}, 1/\sqrt{6}$
- (D) None

Ans: (B)

Q62. Value of $\int e^{(\sin^2 x)} \sin 2x \, dx$ from 0 to $\pi/2$ is:

- (A) e
- (B) e^2
- (C) $e - 1$
- (D) 1

Ans: (C)

Q63. If $f(x)$ is everywhere differentiable, then a, b are:

- (A) $a=3, b=5$
- (B) $a=0, b=5$
- (C) $a=0, b=3$
- (D) $a=b=3$

Ans: (A)

Q64. Origin, z_1 , z_2 form equilateral triangle if roots of $z^2 + az + b = 0$ satisfy:

- (A) $a^2 = b$
- (B) $a^2 = 3b$
- (C) $a^2 = 2b$
- (D) $a = b$

Ans: (B)

Q65. Solution of $x \frac{dy}{dx} + y = x e^x$ is $xy = e^x \phi(x) + C$, where $\phi(x)$ is:

- (A) $x + 1$
- (B) $x - 1$
- (C) x
- (D) x^2

Ans: (B)

Category-III (Q66 - Q75)

Q66. Hyperbola $16x^2 - 3y^2 - 32x + 12y - 44 = 0$ properties:

- (A) Transverse axis $2\sqrt{3}$
- (B) Centre (1, 2)
- (C) Eccentricity $19/3$
- (D) Directrix $x = \sqrt{19/3}$

Ans: (A, B)

Q67. For integral sequence recurrence property:

- (A) a_n in G.P.
- (B) b_n in A.P.
- (C) c_n in H.P.
- (D) a_n in A.P.

Ans: (B, D)

Q68. Let $f:[1,2] \rightarrow \mathbb{R}$ cont and $f'(x) = |f(x)|^2 + 4$. Then:

- (A) $f(3)-f(1)=5$ true
- (B) $f(3)-f(1)=5$ false
- (C) $f(3)-f(1)=7$ false
- (D) $f(3)-f(1)>8$

Ans: (B, C, D)

Q69. If $\phi(x) = f(x) + f(1-x)$ and $f''(x) < 0$ then:

- (A) inc $[0, 1/2]$
- (B) dec $[1/2, 1]$
- (C) inc $[1/2, 1]$
- (D) dec $[0, 1/2]$

Ans: (A, B)

Q70. Vector magnitude and angles property:

- (A) Min $-3/2$
- (B) Max 3
- (C) Can be 0
- (D) Always neg

Ans: (A, B, C)

Q71. Function properties and integral is $2 \log 2$:

- (A) Even
- (B) Continuous
- (C) Differentiable
- (D) $\int f(x)dx = 2 \log 2$

Ans: (B, D)

Q72. Complex equation $|z-w|^2 = |z|^2 - |w|^2$ reps:

- (A) Circle
- (B) Line
- (C) $\operatorname{Re}(zw^*) = |w|^2$
- (D) $z = w$

Ans: (C)

Q73. Function $f(x) = \cos x^2$ is:

- (A) Periodic 2π
- (B) Even
- (C) Continuous
- (D) $f'(0)=0$

Ans: (B, C, D)

Q74. If A, B symmetric matrices:

- (A) $AB+BA$ symm
- (B) $AB-BA$ skew
- (C) $A+B$ symm
- (D) AB always symm

Ans: (A, B, C)

Q75. Line through (a, b) and (-a, -b):

- (A) Passes (0,0)
- (B) Passes (a^2 , ab)
- (C) Slope b/a
- (D) Passes (1,1)

Ans: (A, B, C)